

Fallacies in Academic Writing

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1. Ad-hoc

Definition:

This occurs when someone introduces a specific, often arbitrary, piece of information or "exception" to save their theory from being proven wrong. Instead of adjusting the hypothesis based on new evidence, they invent a reason why the evidence doesn't count.

Example:

"My study proves that all mammals avoid water. Oh, you found a dolphin? Well, dolphins are an exception because they were specially evolved by an ancient underwater civilization to be different from all other mammals."

2. Snowman

Definition:

This involves misrepresenting or oversimplifying an opponent's argument to make it easier to attack. By knocking down this "straw man" (a weak version of the original point), the arguer pretends they have defeated the actual, more complex argument.

Example:

Original Argument:

"We should shift more funding from high-stakes testing toward early childhood literacy programs."

Straw Man Response:

"My opponent wants to completely abolish all educational standards and let children do whatever they want in the classroom."

3. Over-generalization

Definition:

This happens when a broad conclusion is drawn from a sample size that is far too small or unrepresentative. In research, this is often called "sampling bias."

Example:

"I interviewed three students in the library at 6:00 AM, and all of them said they prefer studying in the morning. Therefore, the entire university student body is most productive in the early morning hours."

4. Ad Hominem

Definition:

Instead of addressing the actual argument, the writer attacks the character, background, or traits of the person making the argument.

Example:

"Dr. Smith's theory on carbon emissions is invalid because he was once fired from a previous university for a policy disagreement."

5. Slippery Slope

Definition:

Asserting that a relatively small first step will inevitably lead to a chain of related (usually negative) events without providing evidence for that chain reaction.

Example:

"If we allow students to use AI for brainstorming, they will eventually stop writing entirely, leading to the total collapse of the global education system and a workforce that cannot read."

6. False Dilemma

Definition:

Presenting only two extreme options as the only possibilities when, in fact, a "middle ground" or other alternatives exist.

Example:

"We must either completely ban all fossil fuels immediately or face the certain extinction of the human race by next year."

7. Appeal to Authority

Definition:

Claiming something is true simply because an "authority figure" said it, even if that person isn't an expert in the specific field being discussed.

Example:

"The famous physicist Dr. Hawking said that a plant-based diet is the only way to save the planet, so we should all stop eating meat."

8. Red Herring

Definition:

The arguer doesn't necessarily lie; they simply shift the focus to a separate issue that is easier to defend or more emotionally charged, hoping the audience follows the "new scent" and forgets the original point.

Example:

The researcher begins the research under the topic "Should the university increase its budget for library resources to support graduate research?" but later realizes that it's difficult to defend and writes about allocating the budget to the nearby canteen.

9. Circular Reasoning

Definition:

This occurs when the conclusion of an argument is used as one of the premises. Essentially, you are saying "A is true because B is true, and B is true because A is true."

Example:

A researcher claims that WiFi is a faster technology because the ISP Dialog provides fast WiFi and that Dialog is faster because it has better WiFi without considering other factors.

10. Confirmation Bias

Definition:

Only listing the data that supports the biased result the researcher hopes to get through the research.

Example:

A researcher conducts a survey on AI impact on undergraduate students. However, the researcher is biased towards the positive aspect of AI impact, hence only including the results that prove this point.

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